SAINT MICHAEL COLLEGE OF CARAGA

BARRIERS AND ENABLERS IN ONLINE RESOURCE UTILIZATION AMONG LIBRARY USERS IN SAINT MICHAEL COLLEGE OF CARAGA (SMCC)

A Thesis Proposal Presented To

The Faculty of College of Computing and Information Sciences

In Partial Requirements for the Degree

BACHELOR OF LIBRARY AND INFORMATION SCIENCE

By

Alexander Poriol

Abegail D. Abunda

Jay-uanie M. Binongcasan

Michelle L. Duarte

Mariel C. Ordiz

ADVISER: Lani Jean E. Lumogda

October 2025

TABLE OF CONTENTS

			Pages
	TITLE PAGE		i
	CERTIFICATE OF PANEL APPROVAL		ii
	ABSTRACT		iii
	DEDICATION		iv
	ACKNOWLEDGEMENTS		v
	TABLE OF CONTENTS		vi
	LIST OF TABLES		viii
	LIST OF FIGURES		x
	CHAPTER		
	1	**THE PROBLEM AND ITS SCOPE**	1
		1.1 Background of the study	1
		1.2 Review of Related Literature	2
		1.3 Theoretical Framework	4
		1.4 Schematic Diagram	4
		1.5 Statement of the Problem	5
		1.6 Hypothesis	5
		1.7 Definition of Terms	7
	2	**RESEARCH METHODOLOGY**	9
		2.1 Research Design	9
		2.2 Research Materials and Methods	10
		2.3 Research Environment, Population and Sample	12
		2.4 Research Data and Analysis Plan	13
		2.5 Research Ethical Standards	14
	3	**RESULTS AND DISCUSSIONS**	19
		3.1 Presentation and Analysis of Data	19
		3.2 Proposed Intervention/ Output	19

4		**SUMMARY, CONCLUSIONS AND RECOMMENDATIONS**	44
		4.1 Summary	45
		4.2 Conclusion	46
		4.3 Recommendations	47

REFERENCES	50
CURRICULUM VITAE	59
APPENDICES	
CERTIFICATE OF TECHNOLOGY BASED ASSESSMENT	68

SAINT MICHAEL COLLEGE OF CARAGA
College Of Computing and Information Sciences

CERTIFICATE OF RESEARCH APPROVAL

This thesis entitled **"Barriers and Enablers in Online Resource Utilization among Library Users in Academic Libraries in Agusan del Norte"** presented and submitted by Click here to enter text. in partial fulfillment of the requirements for the degree of Click here to enter text. is hereby accepted and recommended for Oral Examination.

LANI JEAN E. LUMOGDA, RL
Adviser

ABEGAIL D. ABUNDA
Member

ALEXANDER PORIOL
Member

JAY-UANIE M. BINONGCASAN
Member

MICHELLE L. DUARTE
Member

MARIEL C. ORDIZ
Member

PRINCESS VARONA
Chairperson

ACCEPTED as partial fulfillment of the requirements for the degree of Click here to enter text.

APPROVED by the tribunal at the Oral Examination with the grade of _____.

Click here to enter text.
Adviser

Click here to enter text.
Member

Click here to enter text.
Member

Click here to enter text.
Member

Click here to enter text.
Member

Click here to enter text.

Chairperson

SAINT MICHAEL COLLEGE OF CARAGA
Nasipit, Agusan del Norte

College Of Computing and Information Sciences
ABSTRACT

TITLE:	XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX
AUTHOR/S:	XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX
DEGREE:	XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX
ADVISER:	XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX
PLACE OF PUBLICATION:	XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX
DATE	XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX
PAGES	XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX

DELETE BOX AFTER READING

RESEARCH DESIGN -refers To The Specific Approach Use In The Research E.x. DESCRIPTIVE QUALITATIVE

Objectives
Write a short introduction of your paper. Provide the direction of the study and nuances.

Methodology
Provide the specific method that was used in your paper.

Findings
Highlight the significant findings of your paper.

Recommendation
Just highlight the primary recommendation

KEYWORDS: PROVIDE AT LEAST 5

Check guidelines for writing an abstract for further details.

(RESEARCH DESIGN)

I. OBJECTIVES

XXXXXXXXXXXXXXXXXXXXXXXXXXXXX

II. METHODOLOGY

XXXXXXXXXXXXXXXXXXXXXXXXXXXXX

III. FINDINGS

XXXXXXXXXXXXXXXXXXXXXXXXXXXXX

IV. RECOMMENDATIONS

XXXXXXXXXXXXXXXXXXXXXXXXXXXXX

KEYWORDS: XXXXXXXXXXXXXXXXXXXXXXXXXXXXX

DEDICATION

DELETE BOX AFTER READING

Structure of a Dedication
1.**Opening Statement**
•Example: *"This thesis is dedicated to…*
2. **List and Explanation**
•Example: *"To my parents, for their love and support, and to my advisor, Dr. [Name], for invaluable guidance."*
3.**Closing (Optional)**
•Example: *"This work would not have been possible without you."*

ACKNOWLEDGEMENTS

The researchers sincerely extend their deepest gratitude to all individuals and institutions who have contributed to the successful completion of this study, entitled *"Barriers and Enablers in Online Resource Utilization among Library Users in Academic Libraries in Agusan del Norte."*

Foremost, the researchers express their heartfelt appreciation to **Ms. Lani Jean E. Lumogda**, their research adviser, for her invaluable guidance, constructive feedback, and constant encouragement throughout the conduct of this research. Her expertise, patience, and dedication served as a source of inspiration that motivated the researchers to pursue academic excellence and integrity in their work.

The researchers also acknowledge the unwavering support of the **faculty members of the College of Computing and Information Sciences of Saint Michael College of Caraga**, whose insights and mentorship helped refine the study's direction and objectives. Gratitude is likewise extended to the **respondents** who willingly participated in the research survey and provided the data that became the foundation of this study.

Sincere thanks are also offered to the **family and friends** of the researchers for their unending love, understanding, and moral support, which strengthened their perseverance in completing this academic endeavor.

Lastly, the researchers are profoundly thankful to the Almighty God for His guidance, wisdom, and grace that made every step of this journey possible. Without His divine providence, this study would not have been successfully accomplished.

ALEXANDER PORIOL

ABEGAIL D. ABUNDA

JAY-UANIE M. BINONGCASAN

MICHELLE L. DUARTE

MARIEL C. ORDIZ

LIST OF TABLES

Table	Pages
1	
2	
3	

CHAPTER 1

THE PROBLEM AND ITS SCOPE

1.1 Background of the study

The advancement of digital technology has transformed how academic institutions access, deliver, and utilize information. In the 21st century, online resources have become essential for teaching, learning, and research, reshaping the role of academic libraries worldwide. Libraries are transitioning from traditional print collections to hybrid systems that integrate digital databases, e-journals, open-access repositories, and other web-based platforms (Nguyen, 2020; Lu & Lin, 2024). However, the effective utilization of these resources depends on several factors that can either enable or hinder user engagement. Diki (2022) emphasized that utilization is influenced by elements such as awareness, accessibility, technological capability, and user training. Similarly, Mensah (2022) and Musungu and Maina (2023) identified unstable internet connection, inadequate devices, and poor ICT skills as major hindrances to e-resource use across developing regions. Burhansab et al. (2021) and Chandra and Patra (2021) further observed that usability problems and weak promotion often reduce user engagement even when subscriptions and access exist.

On the other hand, various studies underline that library instruction, faculty support, and institutional infrastructure serve as enablers that enhance awareness, satisfaction, and utilization (Abubakar & Nwachukwu, 2021; Rahman & Alam, 2020). Digital transformation studies also suggest that institutional policy and continuous investment in ICT infrastructure are key to turning resource availability into actual usage (Nguyen, 2020; Lu & Lin, 2024). These findings highlight that online resource utilization is a multifactorial issue shaped by behavioral institutional, and technological dimensions, reflecting the importance of understanding both barriers and enablers in academic settings.

In the Philippine context, similar patterns are observed but with challenges unique to local academic environments. Literature documents that many students in the Philippines, particularly in rural and provincial areas, experience limited access to digital tools and connectivity, resulting in underutilization of online library resources (Gonzales, 2021; Santiago & Dela Cruz, 2021; Rosales, 2022). Moreover, insufficient information literacy training has been a

recurring problem among Filipino learners, limiting their ability to locate and use e-resources effectively. However, interventions such as library orientations, faculty integration, and peer mentoring have shown positive results in improving confidence and utilization among students (Abad, 2020; Santos & Villanueva, 2022). While these efforts represent progress, significant disparities persist in provincial institutions where technology infrastructure and awareness campaigns remain unevenly developed.

At Saint Michael College of Caraga (SMCC), a private higher education institution located in Nasipit, Agusan del Norte, Philippines, these challenges are particularly pronounced due to its regional setting and reliance on limited ICT infrastructure. The online library systems at SMCC provide access to the resources/collections which include E-resources, Free/Open Databases, OAR/ OER DIRECTORY, Trial Databases, New Acquisitions, Magzter, and ProQuest which offers dissertations and theses and interdisciplinary materials. The resource access system of the organization experiences multiple access problems which include users encountering login authentication failures and users facing bandwidth restrictions during peak times and users dealing with device compatibility problems which affect their ability to use outdated devices and browsers. The SMCC Learning Resource Center (LRC) serves as the hub for these e-library services, featuring an Online Public Access Catalog (OPAC) for searching physical and digital holdings, a digital repository for institutional publications, and integrated e-library portals that allow remote access via the college's intranet or VPN. These online resources are intended to supplement the print collection and align with SMCC's outcomes-based and technology-supported learning initiatives.

The online library resources at SMCC experience low usage according to librarian reports and internal usage records. Students continue to use free internet sources instead of the LRC's scholarly databases despite the availability of existing subscriptions and e-library services. The librarians found that users experience multiple challenges when they try to access databases which include their lack of database knowledge ongoing login credential problems and their need for internet connection and information literacy skills. The policies of the library restrict

on-campus access to some resources which prevents students who need to study outside their designated time from using those resources. The observed problems show that SMCC students need more than access to online library resources to effectively use these resources.

Although these studies provide valuable insights, a significant research gap remains. Most prior works tend to isolate either barriers or enablers in analyzing online resource utilization, rarely addressing both simultaneously. Consequently, limited evidence exists on how these factors interact and collectively influence user behavior, especially in institutions like Saint Michael College of Caraga (SMCC). Studies by Panhilason (2024a) and Azrael and Florentino (2025) have examined user awareness and satisfaction with online library resources but failed to fully account for the combined effects of infrastructural, behavioral, and instructional variables. Similarly, reports from St. Peter's College of Toril, Inc. (2025) show that high satisfaction does not always equate to frequent usage, suggesting that hidden barriers such as poor device access, lack of awareness, or limited institutional support persist. Addressing these gaps is essential to developing a comprehensive understanding of how barriers and enablers collectively affect online resource utilization among library users.

Given these gaps, this study titled "Barriers and Enablers in Online Resource Utilization among Library Users in Saint Michael College of Caraga (SMCC)" aims to examine the key factors that hinder or promote the use of online library resources among students. It seeks to identify the most common barriers to utilization, recognize the enabling factors that encourage engagement with online resources, and determine the relationship between these factors and the extent of online resource use. The study is anchored on the Unified Theory of Acceptance and Use of Technology (UTAUT), which explain how perceived usefulness, ease of use, and facilitating conditions influence users' decisions to adopt and utilize technology.

The findings of this study will provide practical and evidence-based insights for multiple stakeholders at SMCC. For students, the results will help them identify challenges that hinder effective utilization of online resources and recognize enabling factors that improve access, confidence, and academic performance. Librarians and faculty can use the findings to design

instructional programs, library orientations, and promotional strategies to increase engagement with e-resources. Academic administrators can leverage the results to inform policy-making, budget allocation, and ICT infrastructure improvements to ensure equitable digital access.

Moreover, this study contributes to the digital transformation of SMCC's library services by providing measurable evidence on barriers, enablers, and utilization levels, supporting data-driven strategies to improve technology adoption. By examining both enabling and hindering factors quantitatively, the research ensures that interventions are grounded in empirical data, enhancing the library's role in fostering academic success and lifelong learning. Ultimately, the study underscores the importance of integrating technological, behavioral, and institutional considerations to maximize online resource utilization, aligning with the principles of inclusive and effective education.

1.2 Review of Related Literature

The rapid development of scholarly libraries into hybrid and digital spaces has redefined the way users access, process, and utilize information materials. The incorporation of web resources into library systems is critical to support research, instruction, and lifelong learning. However, despite global efforts toward digitalization, disparities still exist in the way users access and utilize these resources, particularly in developing regions. Libraries in the Philippines continue to face a mix of infrastructural, pedagogical, and behavioral challenges that affect the viability of full-digital-platform usage.

This chapter presents a literature review that investigates both the barriers and enablers in online resource utilization among academic library users, guided by the variables presented in the study's conceptual framework. The discussion is organized into four major parts:

(1) Online Resources, which introduces the concept and scope of digital learning materials;

(2) Barriers to Online Resource Utilization, focusing on awareness gaps, technological limitations, and resource availability;

(3) Enablers to Online Resource Utilization, emphasizing librarian and faculty support, user training, and adequate infrastructure; and
(4) Trends in Online Resource Utilization, including frequency of use, purpose, satisfaction, and effectiveness.

Each section synthesizes both foreign and local studies to identify research gaps and contextual relevance of this study within Agusan del Norte.

Online Resources

Online resources are digital information materials accessible through the internet that support academic, instructional, and research activities. These include electronic journals, e-books, online databases, and open-access repositories that provide users with quick and reliable access to scholarly content. According to Nguyen (2020), online resources have become the backbone of modern academic libraries, offering flexibility and broader information reach beyond physical collections. Similarly, Al-Khatib et al. (2024) emphasized that digital platforms enhance learning efficiency by allowing students and faculty to retrieve updated materials anytime and anywhere.

In the Philippine setting, Lucero et al. (2021) highlighted that integrating online resources into university libraries significantly strengthens teaching and research productivity, although technical and infrastructural limitations continue to affect their full utilization. Estrada et al. (2023) also found that consistent access to digital resources improves students' academic readiness and promotes self-directed learning. Moreover, Cabrera (2024) noted that effective use of these resources depends on adequate user training, librarian assistance, and institutional support.

Overall, online resources serve as the foundation of digital learning environments, promoting accessibility, efficiency, and innovation in academic research. Their effectiveness,

however, relies on users' digital literacy, awareness, and the presence of supportive technology factors further examined in the succeeding sections.

Barriers to Online Resource Utilization

Lack of Awareness

Awareness in digital library resources plays a vital role in determining how effectively users can utilize online information resources. According to Rosales (2024), students at universities in Iloilo City demonstrate moderate knowledge and occasional use of electronic resources, which are often employed for research activities. However, many still lack the skills to navigate online databases, which indicates limited participation in technology-based learning systems. The study indicated that only awareness was inadequate; Practical training, continuous promotion, and integration of digital resources are required in academic activities to increase the level of practical use.

In a similar study, Kuldeep Singh (2019) reviewed students and scholars at Punjabi University, Pataala, and he found that while many users were aware of electronic resources such as e-journals and E-books, slow internet connection remains a hindrance to the internet, poor infrastructure, and limited time of operation. Singh noted that although the users' levels of satisfaction with digital access, the lack of continuous awareness programs and inadequate ICT infrastructure continue to prevent their full use. The study suggested the implementation of continuous awareness campaigns and skill enhancement workshops to fill the gap between consciousness and actual use of online resources.

Meanwhile, Humbhi et al. (2023) at the University of Balochistan, the level of knowledge and use of e-resources of teachers is higher than students. The study identified some major obstacles such as frequent power outages, unstable Internet connections, and lack of proper training sessions. These limitations, in conjunction with the lack of promotional initiatives, have caused users to fully maximize the potential of e-resources. The authors have suggested the greater investment of institutions in awareness-building programs, improving ICT facilities, and

implementing professional development initiatives to ensure equal access between different user groups.

Palacay (2025) also emphasized that lack of awareness is related to greater infrastructure and institutional barriers. The study has shown that old and outdated materials, lack of support from librarians, and time deficiency limits the ability of teachers and students to use digital library systems effectively. The author mentioned that in order to address such obstacles, academic institutions need to implement proactive outreach strategies, modernization of library collections, and integrate awareness initiatives in the very culture of the academy through the continuous participation of librarians.

In addition, Jameson et al (2019) discovered. Although students appreciate the help of librarians, most people still prefer to find information alone and rarely seek professional help. The main reason for this is the limited awareness of the reference services offered by the library. The study indicated that the implementation of awareness campaigns and digital marketing strategies adapted to students' online behavior would help improve their engagement and use of library resources.

Meanwhile, in Hazarika's (2024) study of college students in Upper Assam, 61% of participants were unaware of electronic resources and still relied on traditional materials. Among the major obstacles mentioned by the study are the lack of ICT infrastructure, low levels of digital literacy, and limited participation of librarians. Research emphasized that strengthening awareness requires a holistic approach that combines more enhanced ICT access, users' training (user training), and simplified interface to make the use of electronic resources easier and more inclusive.

Similarly, Florentino (2025) examines the level of awareness and validity of the use of Open Access Resources (OARS) to students of the Lyceum of the Philippines University. The study found that although students considered the oars useful, many of them had difficulty finding and using full-text documents due to navigation problems. The author suggested implementing the

structured marketing plan, enhancement of interface design, and providing user-oriented training to expand the effective use of open access resources. In sum, the literature emphasizes that awareness serves as the foundation of the use of resources, but if it does not include technical skills or infrastructure support, it will only have a limited impact.

As such, continuous training programs, librarian-led orientations, and targeted promotional strategies are required to continue the transition from simple awareness to active participation and use of online resources.

Technological Limitations (Poor Internet, Low ICT Skills)

Technological limitations remain among the most common challenges in using online resources at various academic institutions. These obstacles include unreliable internet connection, old or obsolete digital infrastructure, technical support deficiency, and low -level digital literacy of users. According to Shikali et al. (2022), during the Covid-19 Pandemic, Academic Libraries in Africa conducted various digital solutions such as Virtual Reference Services and Off-campus access systems. However, weak and unstable Internet connections, limited ICT devices, and low digital literacy, which are a major obstacle to the effectiveness of these initiatives have remained a major problem. The authors pointed out that sustainable technology (Sustainable Technological Investment) and fairly accessible Internet access (Equitable Internet Access) are required to ensure continuous and inclusive participation in the academic field.

In the context of the Philippines, Lucero et al. (2021) both public and private institutions of higher education show relatively high levels of e-learning readiness. However, they are still experiencing problems such as infrastructure deficiency, slow internet connections, and limited access to timely technologies. The study found that these technological limits impede the successful integration of e-learning platforms, although both teachers and students are willing to adapt to them. The authors have suggested improving ICT infrastructure, regularly providing

technical training, and establishing institutional levels that support digital transformation in the field of education.

In the same way, Alasa et al. (2022) emphasized that polytechnic students in Nigeria often use search engines and internet facilities for academic purposes. However, their access to E-Journals and E-Books remains limited due to unstable power supply, slow Internet connections, and lack of personal devices. The study has shown that while students' strong awareness and motivation, technical and infrastructural issues remain a hindrance to the effective use of online resources. The authors emphasized that investing in ICT infrastructure, stable power supply, and student-based school programs are important to maximize online resources.

Meanwhile, Fabito et al. (2021) conducted a study of digital challenges experienced by students taking computing courses in the Philippines during the Covid-19 Pandemic period. Their findings have identified three main obstacles: unreliable internet connection, not suitable home education areas, and lack of teachers' readiness for online instruction. According to the study, these problems affect not only access to learning materials, but also students' overall academic participation. Researchers have suggested that technology barriers are not only infrastructural but also pedagogical, so teachers should focus on the readiness and support policy from the institution.

In Uganda, Bruuga et al. (2023) found that universities in Uganda continue to face major challenges in the use of electronic information resources. The study identified unreliable Internet connections, lack of ICT infrastructure, and limited use of users as common problems. Through swot analysis that includes students and teachers, the study emphasized that although there are e-resources, technical problems such as server downtime, power interruptions, and limited bandwidth still impede effective use. The authors have suggested increased investment in ICT infrastructure and proper fund management to ensure continuous technological development and fair access.

In addition, Durodolu et al. (2024) mentioned that poor Internet connection directly affects the productivity and satisfaction of academic library staff. Participants reported the decline in cataloging, data retrieval, and virtual service delivery due to slow or unstable connections. The study emphasized that sustainable use of digital resources requires not only access to students but also institutional support for the technological needs of librarians and library staff.

In total, these studies emphasize that technology-related barriers-specifically poor Internet connections, ICT infrastructure deficiency, and limited digital literacy-directly impede the full use of online library resources. Effective digital access depends on stable technological environments, adequate training with users, and continuous institutional investment in infrastructure and capacity-building.

Limited Resource Availability

Limited availability of resources is another important barrier to the effective use of online resources in academic libraries. These include lack of timely materials, limited access to subscription databases, lack of funding for new libraries or digital acquisitions, and lacking ICT support. According to Mwantimwa et al. (2021), the gap between consciousness and actual use of e-resources at Tanzania universities is caused by limited ICT infrastructure and unreliable Internet services. The study found that lack of technological facilities and lack of adequate training of librarians prevent academic productivity. The authors have suggested cooperation at the government level to improve infrastructure and promote resource-sharing between institutions.

In Pakistan, Khalid Usman et al. (2023) identified Similar challenges among distance learners, including limited ICT skills, lack of guidance from institution, and inadequate training of library staff. The authors have noted that the weak quality of library websites, poor internet signals, and the absence of marketing initiatives have increased the interest and use of users in

library services. The study suggested developing orientation and enhancement of digital collections to increase users' confidence and active participation.

Meanwhile, Norch et al. (2022) found that despite high consciousness levels, actual use is not regular due to retrieval difficulties, limited off-campus access, and inconsistent ICT support. The study determined that periodic training and easily accessible online tutorials can greatly contribute to the improvement of search competencies and more consistent use of e-resources

In the Philippine context, Roman et al. (2020) emphasized that although students recognize the importance and benefits of e-resources, low levels of awareness and technical skills are hindering their effective use. The authors also mentioned the importance of regular orientation and cooperation by teachers to integrate online resources into coursework, which will help encourage continuous use and higher levels of student engagement.

Raphael Aryee et al. (2024) explored the behavioral consequences of limited resources, which discovered that the lack of materials in universities in Ghana has caused frustration among users and has led to negative coping mechanisms, such as hiding or simplicity in library materials. These findings show the social and ethical dimensions of resources deficiency, which is associated with the moral decline of users and institutional efficiency.

Meanwhile, Estrada et al. (2023) found a positive relationship between frequent library use and academic success, emphasizing that regular access to timely materials contributes to higher levels of academic performance. The study suggested expanding E-Library access and investment in further training librarians to address both technological and resource-based barriers.

Finally, Papay et al. (2023) examined the effectiveness of Blended Library Services at the University of Baguio and discovered that the integration of traditional and online services has contributed greatly to increasing user engagement. However, some services are still less available

due to limited users' awareness and lack of integration with academic programs. The study indicated that expanding blended models and strengthening promotion can help maintain the use of resources after pandemic.

Together, these findings indicate that limited resources are not only a logistical issue but a systemic barrier that is shaped by funding, institution policy, and users' training. Effective use of online resources requires a careful balance between expanding resources, equal access, and institutional support for continuous development.

Accessibility Issues

Accessibility issues cover both physical and digital barriers that prevent users from using online resources effectively. According to Sherman (2022), in collaboration with the American Library Association, deficiencies such as inadequate funding for assistive technologies, lack of training of staff, and low awareness of the needs of the disabled. As such, it results in library platforms and digital collections that are not fully accessible for users with physical or cognitive limitations. Similarly, Comeaux and Schmetzke (2013) and Vaughan and Warlick (2020) have found that many academic library websites do not comply with accessibility standards, so students with visual disabilities or thinking are difficult to navigate.

Higgins (2013) emphasized that marginalized populations, including rural areas and low-income, continue to experience obstacles to access to limited Internet connections, digital literacy deficiency, and lack of local content. The study also showed that even with access, the usability issues and old systems continue to affect users' participation. It strengthens the digital divide and emphasizes the need for more inclusive strategies in academic libraries.

Hamid et al. (2024) discovered that infrastructure deficiency, low levels of digital literacy, and high Internet costs prevent the access of staff and students in online resources in Nigeria colleges. They have suggested developing institutional training and support to overcome technological inequality. Similarly, Obande et al. (2020) that poor Internet connection, ICT skill

deficiency, and unstable power supply hindered the full use of e-resources to universities in Nasarawa State.

Lawal (2019) also noted that despite sufficient resources in agricultural universities in Nigeria, students are still experiencing intense accessibility issues, such as difficulty finding materials and lack of knowledge of the use of search techniques. Pope and Thompson-Franklin (2025) adopted it, saying that the challenges of accessibility also included poor promotion of services and limited collaboration between librarians and faculty members. They emphasize the need for proactive institutional framework for more inclusive access to digital resources.

Finally, Okolo (2024) confirmed that the lack of ICT infrastructure and unstable Internet connection remains a major obstacle to the academic libraries of the 21st century. The study found that limited access to subscription databases and lack of adequate training of students and staff is preventing the full use of e-resources. Okolo noted that without the systematic investment in users' infrastructure and education, accessibility barriers will remain despite the continued development of technology.

Enablers to Online Resource Utilization

Although academic libraries continue to experience various barriers such as infrastructure deficiency, limited awareness, and accessibility issues, some studies have identified important factors that enhance the use of online resources. These include the support and guidance of librarians, collaborations of teachers and staff, users' training, and provision of adequate infrastructure. In sum, these factors shape users' attitude, ability, and motivation to actively interact with digital information sources. The integration of these enablers in institutional systems ensures that students, teachers, and staff will greatly benefit online resources as support for teaching, learning, and research.

Librarian Support and Instruction

The help of librarians is recognized as one of the most important factors in enhancing users' participation in online resources. The role of librarians is important in guiding students so that they can be good at finding information, critical to reviewing resources, and responsible for using information. According to Okolo (2024), the proper structure of the librarian support-such as orientations, guided database navigation, and personalized assistance-has greatly contributed to develop students' ability to use e-resources effectively. However, Okolo also noted that fundraising and poor ICT infrastructure are a hindrance to the librarian's ability to provide continuous service. The study suggested that strong support from the institution and ongoing professional development of librarians were needed to maintain the active use of online resources.

Meanwhile, Hashim, Shuhidan, Anwar, and Yunus (2022) have discovered that the use of postgraduate students in the web-based library and information services in academic institutions in Malaysia is strongly affected by the quality of library technology, support, and environment. When librarians are approachable, well -trained, and with adequate technical expertise, students' levels of engagement and satisfaction are higher in using online databases. The study suggested the strengthening of librarians and the integration of digital support systems to increase users' confidence and skills.

Isah, Abdullahi, and Muhammad (2019) pointed out that structured library instruction programs such as orientations and classroom-based information literacy sessions help students develop their skills in independent research. The study revealed that freshmen who attended the library orientations had a higher level of database navigation and resource evaluation skills compared to those who did not attend. The authors emphasize that continuing teaching led by librarians promotes lifelong learning and critical information management of students.

Rasmussen and Sørensen (2024) pointed out that effective teaching of digital literacy is one of the major enablers in using online resources. In their study at the University of Copenhagen Library, it was discovered that when librarians actively participate in courses and cooperate with teachers, students are becoming more efficient in searching, evaluating, and

using digital information. This model of teamwork, called the embedded librarianship, increases students' participation and confidence in the use of online library systems. The study shows that the close relationship between librarians and teachers is an important enabler for the more effective use of online resources in academic institutions.

These findings have jointly proved that the support and teaching of librarians not only increases awareness, but also deepens the participation and critical thinking of users. When librarians have adequate training, proper infrastructure, and strong support from the institution, they become the main partner in the development of the effective use of online resources in academic institutions.

Faculty Support

Faculty involvement participation is also an important enabler that connects classroom teaching and the use of library resources. Teachers play a major role in encouraging students to explore digital resources, use them in their academic activities, and cultivate independent research skills. Chukwuma and Okafor (2022) pointed out that teachers serve as a bridge between students and libraries, and that their active participation determines how effective students are to use in online materials. The study found that when teachers incorporated e-resources into their courses and encouraged students to use library databases, students' levels of participation and academic performance were significantly raised.

Meanwhile, Morris and Currie (2019) reported that the lack of collaboration between teachers and librarians often leads to low use of online resources. The study determined that when professors do not include library links or resource lists in their teaching materials, students are not aware of available databases. Researchers have recommended to strengthen teachers and librarians as a strategy to integrate information literacy into the curriculum.

Becksford (2022) noted that the poor relationship between librarians and teachers limits collaboration with the design of instructional content. Research pointed out that many teachers do not fully recognize the important role of librarians in teaching, causing the opportunity to

integrate online resources into academic activities. The study suggested that teachers and librarians and librarians could intensify and relationships to overcome such challenges and improve the accessibility of digital resources.

Meanwhile, Nguyen and Tuamsuk (2020) conducted a study on the collaboration of teachers and librarians at Vietnam universities. They have found that poor infrastructure and lack of institutional coordination has pushed professors to rely on external sources of information rather than the use of institutional databases. This situation has shown that collaboration and old or outdated library systems negatively affect the quality of academic research and teaching.

Similarly, Spangler and Casper (2023) found that teachers who cooperated with embedded librarians have experienced major improvements in students' motivation, information literacy, and confidence in using online resources. However, the study also emphasized that the inconsistent or poor communication and lack of knowledge about the capabilities of librarians impedes the long-term success of this type of collaboration.

Moreover, Faix and Baker (2022) said that digital tools like Leganto help teachers more effectively choose and share digital materials, but technical problems, license limits, and time deficiency are a hindrance to its broader use. The study emphasizes the importance of simplifying resource-sharing systems and providing faculty development programs to facilitate teachers' access to online resources.

Meanwhile, Luke (2011) emphasized that faculty in-service programs are important in disseminating knowledge of library services and resources. When professors have enough information about the extent of available resources, they become more effective in guiding students to related databases and digital tools. Thus, the continued participation of teachers strengthens the relationship between instruction and information access, which strengthens the overall learning system.

Collectively, these studies show that strong collaboration between teachers and librarians enhances students' ability to search, evaluate, and use digital information effectively.

Teachers 'support serves as a motivational and structural enabler that ensures not only the students' access to online resources, but also their significant use of their academic journey.

User Training

The training of library users is a major factor that provides them with the technical and analytic capabilities needed for the effective use of online resources. Success programs help to boost users' confidence, reduce search errors, and improve navigation efficiency in digital databases. Adebayo (2021) pointed out that integration of teaching about digital sources in the classroom enhances students' ability to identify and evaluate academic materials. The study also showed that the combined training of librarians and teachers results in higher students' participation and more effectively developing their Information Literacy skills.

Also, Fernández-Ramos (2019) defined the lack of cooperation of teachers as a major obstacle to the success of online information literacy instruction at universities in Mexico. The study suggested the collective efforts of librarians and teachers in designing training modules in accordance with the purposes of the curriculum. Through this, students better understand the importance of online resources in their academic needs, resulting in continuous use and better performance.

According to Isah et al. (2019), orientation sessions in the library for new students greatly contribute to students' understanding of the use of resources, citation administration, and navigating databases. Regular training, especially in the early years of college, helps to develop a strong foundation of self-study skills and effective use of resources.

Hashim et al. (2022) and Okolo (2024) emphasized that continuous training and refresher courses are essentially carried out due to the rapid change of digital tools and platforms. With the development of technology, seasonal workshops help keep users conscious and accustomed to new interfaces, new capabilities, and database systems. In addition, training programs that are

in line with the specific needs of users-such as subject-specific orientations-deepen academic participation and increase the level of satisfaction of users.

Effective training programs for users not only expand individual skills but also promote the culture of information literacy within institutions. It also promotes equal access to knowledge and the critical review of information, which in accordance with the global standard of digital education.

Adequate Infrastructure

Adequate infrastructure serves as the foundation of all other factors that practice the use of online resources. Without a stable Internet connection, proper hardware, and innovative software systems, access to digital resources will remain limited. Okolo (2024) pointed out that fundraising, unstable electricity supply, and the use of old or outdated technology prevents the operation of libraries and the productivity of staff. The study emphasized that infrastructure development should be considered as a strategic priority to improve the accessibility and efficiency of the service.

Also, Chukwuma and Okafor (2022) reported that non-continuous internet services and lack of ICT infrastructure reduced the validity of the use of e-resources in academic institutions. The authors have suggested the comprehensive investment of institutions in network expanding, consolidation of energy supply, and technology modernization.

Hashim et al (2022) also mentioned that a supporting technological environment - with adequate bandwidth, reliable servers, and properly maintained computer facilities - directly increases users' satisfaction. Institutions that continue to upgrade their systems and provide prompt technical support reports increased participation in digital platforms.

Infrastructure is not only limited to physical equipment but also includes digital architecture, such as cloud-based library systems and online authentication platforms. When properly implemented, these technologies ensure continuous and safe access to academic

materials. Therefore, sustainable infrastructure investment serves as the main foundation of the library's effective service, which enables other factors such as training, support, and collaboration that operate optimal.

Online Resource Utilization

The use of online resources represents a collective process by which access to, deepens the interaction, and users apply electronic information for academic and research purposes. The extent and method of use depends on user awareness, digital literacy, institutional infrastructure, and personal motivation.

In academic libraries, online resources such as electronic journals, databases, e-books, and digital repositories serve as important tools for enhancing learning, teaching, and research productivity. Many dimensions set the pattern of use, including access frequency, purpose of use, level of satisfaction, and perceived effectiveness. Understanding these dimensions provides an insight into how libraries can optimize their services to meet the constant changing needs of users in digital information.

Frequency of Access

The frequency of access measures how often users participate in online resources within a particular period. Regular use usually shows both familiarity and perceived value of available materials. According to Estrada et al. (2023), the continuous use of library resources has a strong positive relationship with the academic performance of undergraduates in the Philippines. Their study has shown that students who use digital resources more often have achieved higher academic averages and have shown better courses in coursework and exams. These findings emphasize that the frequency of access serves as an indication of the academic participation and the importance of the library.

On the other hand, Norch et al. (2022) stated that although postgraduate students in Ghana are highly knowledgeable about e-resources, their use is not regular. The study associates

irregular use of time deficiency, limited ICT facilities, and heavy off-campus access limitations. The authors have suggested that improving online authentication systems and extension of virtual access hours will help make participation in library databases more consistent.

Shikali et al. (2022) also reported that during the Covid-19 period, many academic libraries in Africa experience fluctuating access patterns due to unstable internet and lack of devices. Despite expanding remote access initiatives, many students do not have enough infrastructure to maintain regular use. Researchers conclude that the consistent frequency of access not only depends on knowledge, but is deeply linked to infrastructure and socio-economic conditions.

The findings indicate that the high frequency of using online resources is reaching through a combination of factors such as awareness, technological readiness, and institutional support (institutional support). Libraries that monitor usage data and adapt their services based on the pattern of using students can promote sustainable engagement and enhance their digital platforms to be more effective in academic context.

Purpose of Use (Assignments, Research, and Self-Study)

Purpose of Use refers to the main purpose or motivation behind students' interaction with online resources. Whether it is for conducting assignments, research, or self-study, the purpose of the goal and quality of use has a significant influence. According to Briones et al. (2023), Filipino learners who deliberately use online resources for academic activities have shown higher self-regulation and digital engagement. Students who use e-learning tools for research and coursework have shown better adaptability and focus, supporting the view that purposeful engagement enhances learning outcomes.

Similarly, Defrain et al., (2025) emphasized that when students use the academic research guides provided by the libraries, they improve their search capabilities, evaluation, and the combination of scholarly information. These guides, often integrated with institutional

websites, strengthen the purpose-driven use of online tools and result in more effective academic work.

Agusiobo (2025) also featured the role of institutional support in encouraging the purposeful use of digital platforms. The study has shown that when educational governments promote structured digital environments, students use online resources for assignments and research. In this way, administrative guidance changes digital engagement from passive browsing to purposeful learning.

In addition, Susanti et al. (2021) stated that students who use self-study applications such as learningapp.org have developed stronger independent learning habits. In addition, Patil et al. (2024) discovered that students who use technology-enhanced self-study tools are more successful in the end of the assignments and have a deeper understanding of the concept. The same study states that purposeful digital engagement improves the total effectiveness of using online resources.

Tong et al. (2022) mentioned that in the blended learning environment, students directly accessing online materials for assignments and research are performing better than its users suddenly.

Similarly, Toroba and Tagadias (2024) discovered that self-regulation is the dedicated relationship of digital resource use and academic success, showing that disciplined and goal-oriented use of online resources resulting in optimal academic results.

These findings suggest that the academic purpose behind online resource use determines the depth of learning and retention. Purposeful engagement whether for research, assignments, or self-study transforms online access into meaningful academic development.

Level of Satisfaction

User Satisfaction represents the perceived quality and amount derived from the use of online resources. It shows the effectiveness of the library service, system usability, and user experience. Lawal and Kannan (2020) have found that student satisfaction is greatly influenced by the availability, accessibility, and quality of library resources. Their study showed a strong relationship between adequate facilities and positive outlook of students in the library effectiveness. Similarly, Abuhassna et al. (2020) that students who use properly design online platforms have experienced higher levels of satisfaction due to user-friendly systems, interactive features, and continuous access to materials.

Maryam al-Khatib et al. (2024) identified four major factors that affect student satisfaction with online learning: Instructional Design, Interaction quality, technology reliability, and student support. Their research emphasized that digital platforms centered on the user promote both satisfaction and performance.

In the context of the Philippines, Teofilo-Labitad and Clarity (2025) confirmed that the availability of digital resources directly increases students' motivation and satisfaction, as easily accessible materials promote more active and purposeful learning.

Similarly, ofordile et al. (2019) stated that students who often use E-library databases report increased satisfaction due to their easy access, relevance, and user-friendly. On the other hand, Panhilason (2024) emphasized that lack of knowledge lowers the level of satisfaction, as unknown users find it difficult to effectively navigate online resources. The study advised awareness campaigns and information literacy sessions to improve enjoyment and engagement.

Overall , satisfaction appears as an important measure of library efficiency. When users feel that digital systems are reliable, relevant, and supporting their academic goals, their satisfaction increases, resulting in ongoing use and improved learning results.

Effectiveness of Utilization

The efficiency of use reflects the extent to which users' participation in online resources contributes to academic success and knowledge growth. It covers the efficiency, relevance, and ability of productive information available. According to Olofinkua et al. (2025), the effective use of digital resources enhances the performance of academic staff and supports teaching and research. However, their study emphasized that limited awareness and technological barriers reduce total efficiency. Similarly, Daramola (2023) emphasized that students' participation in E-Resources at the Nigeria's National Open University expanded access to academic information but was affected by diversity in digital literacy.

Platino (2023) emphasized that although digital libraries have great potential, many students show only moderate participation, which limits learning benefits. Similarly, Rosales (2024) discovered that deficiency in digital training prevents students' ability to take care of e-resources greatly despite their existence. The same study indicates that training and continuous exposure to digital tools are important conditions for effective use.

Basar et al. (2021) found that technological limitations, particularly poor Internet connections and motivation deficiency, impede effective use. Their findings have shown that stable learning environment and supporting digital tools enhance the efficiency of use. Similarly, Medicherla et al (2024) confirmed that students who enjoy the service of the library also show increased effectiveness in the use of information, indicating a positive relationship between satisfaction and academic performance.

Adeyeye et al. (2022) found that at Covenant University, students who regularly used digital platforms such as Moodle and Zoom achieved improved academic performance. The study established a direct relationship between sustained engagement and academic achievement, underscoring that effective utilization transforms access into measurable educational outcomes.

Taken together, these studies highlight that effectiveness is not determined by access alone but by informed, motivated, and consistent engagement. Continuous exposure, technical

competence, and institutional support collectively ensure that digital resource utilization translates into academic productivity and lifelong learning.

This review of related literature explored both the barriers and enablers influencing online resource utilization in academic libraries, alongside the behavioral patterns of usage. The studies collectively revealed that while challenges such as limited awareness, technological limitations, and accessibility issues hinder engagement, strong enablers including librarian support, faculty collaboration, user training, and adequate infrastructure serve as vital mechanisms for promoting effective utilization.

Furthermore, thematic analysis across various dimensions frequency of access, purpose of use, level of satisfaction, and effectiveness demonstrated that online resource utilization is multifaceted. Purpose-driven engagement enhances learning outcomes, satisfaction fosters continuity of use, and effective utilization directly correlates with academic success.

In conclusion, the literature underscores that the success of online resource utilization depends on a balanced ecosystem where institutional infrastructure, user competence, and pedagogical collaboration work together. Strengthening these interconnected factors ensures that academic libraries can fulfill their evolving role as digital learning hubs, bridging the gap between technology and education to support student success and institutional growth.

Synthesis

The reviewed foreign studies emphasize that effective utilization of online library resources is strongly influenced by users' technological readiness, perceived usefulness, institutional support, and availability of digital infrastructure. International research highlights the role of well-developed digital library systems, consistent user training, and strong integration of online resources into academic instruction in promoting high levels of utilization. These studies demonstrate that when users perceive online resources as useful,

easy to use, and institutionally supported, adoption and sustained usage significantly increase.

Similarly, local studies conducted in Philippine academic institutions reveal comparable challenges and opportunities in online resource utilization. While online databases and e-library services are increasingly available, utilization remains uneven due to barriers such as limited awareness, insufficient ICT skills, unstable internet connectivity, and lack of systematic user orientation. Local literature further underscores the importance of librarian assistance, faculty encouragement, and institutional policies in improving students' engagement with online library resources.

Despite the relevance of these findings, most existing studies focus on large universities or urban-based institutions, with limited attention given to provincial and private higher education institutions. In the context of Saint Michael College of Caraga (SMCC), where online library resources are available through the Learning Resource Center, there is limited empirical evidence examining how specific barriers and enablers influence actual utilization among library users. This gap highlights the need to contextualize global and local findings within the SMCC setting.

Thus, the present study draws from both foreign and local literature to examine the barriers and enablers affecting online resource utilization among library users at SMCC. By grounding the investigation in existing studies while focusing on an institutional context that remains underexplored, this research aims to provide data-driven insights that can support evidence-based improvements in library services, digital access, and user support mechanisms at SMCC.

1.3 Theoretical Framework

This study is anchored on the Unified Theory of Acceptance and Use of Technology (UTAUT) by Venkatesh et al. (2003), which serves as the theoretical lens in examining the barriers and enablers affecting online resource utilization among library users at Saint Michael College of Caraga (SMCC). The UTAUT framework is used to explain why and how users decide to accept, adopt, and continuously use online library resources within an academic environment.

The UTAUT model proposes that technology usage is influenced by four key constructs: Performance Expectancy, Effort Expectancy, Social Influence, and Facilitating Conditions. In this study, these constructs provide the theoretical grounding for understanding online resource utilization behavior among SMCC library users.

Performance Expectancy explains users' belief that online library resources contribute to improved academic performance, such as faster access to scholarly materials, improved research quality, and academic efficiency. Effort Expectancy explains users' perception of ease or difficulty in using online library platforms, including database navigation, login procedures, and search tools. Social Influence reflects the extent to which instructors, librarians, and peers encourage or motivate students to utilize online library resources. Facilitating Conditions refer to the availability of institutional support, technological infrastructure, internet connectivity, and user training provided by SMCC.

Within this theoretical lens, barriers such as limited ICT skills, lack of awareness of available online resources, unstable internet connectivity, and inadequate institutional support negatively affect Effort Expectancy, Performance Expectancy, and Facilitating Conditions, resulting in lower levels of online resource utilization. On the other hand, enablers such as librarian assistance, faculty encouragement, user orientation programs, and accessible digital platforms strengthen these UTAUT constructs, thereby increasing users' intention to use and actual utilization of online library resources.

By applying UTAUT as the guiding framework, this study conceptualizes online resource utilization as a result of the interaction between individual perceptions, social influence, and institutional readiness. This theoretical grounding allows the study to systematically analyze how identified barriers and enablers influence technology acceptance and usage among library users in SMCC, providing a strong theoretical foundation for the research variables and the overall study design.

1.4 Schematic Diagram

Independent Variables **Dependent Varisbles**

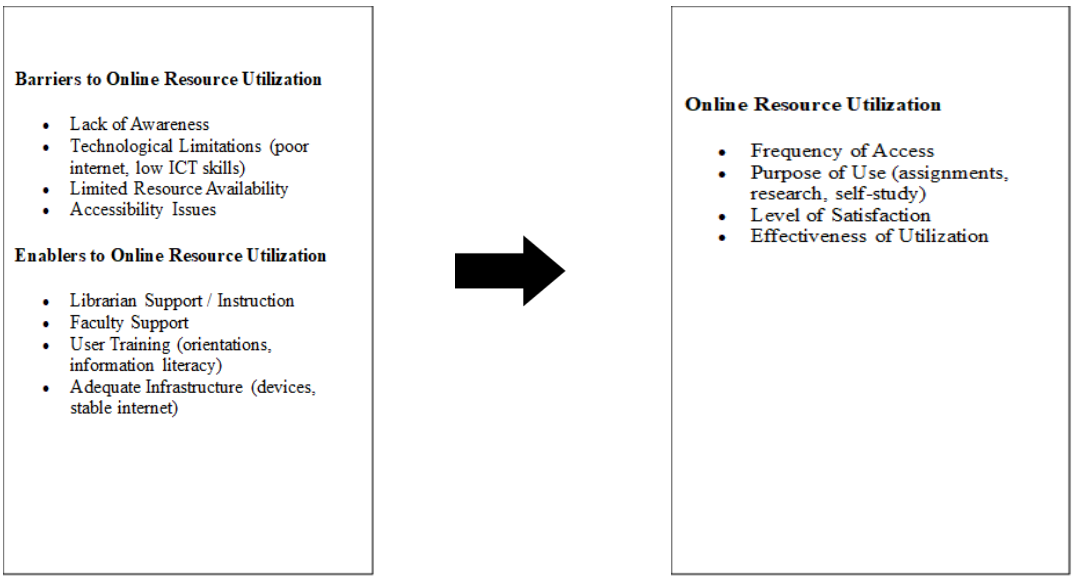


Figure 1. This Schematic Diagram shows the relationship between the independent and dependent variables of the study.

1.5 Statement of the Problem/ Objectives of the Study

This study aims to determine the extent of barriers and enablers in online resource utilization among library users at Saint Michael College of Caraga (SMCC). Specifically, it seeks to answer the following questions:

1. What is the extent of barriers that affect the utilization of online resources among library users at SMCC in terms of:

1.1 lack of awareness,

1.2 technological limitations,

1.3 limited resource availability, and

1.4 accessibility issues?

2. What is the extent of enablers that support the utilization of online resources among library users at SMCC in terms of:

2.1 librarian support and instruction,

2.2 faculty support,

2.3 user training, and

2.4 adequate infrastructure?

3. What is the level of online resource utilization among library users at SMCC in terms of:

3.1 frequency of access,

3.2 purpose of use,

3.3 level of satisfaction, and

3.4 perceived effectiveness?

4. Is there a significant relationship between the identified barriers and the level of online resource utilization among library users at SMCC?

5. Is there a significant relationship between the identified enablers and the level of online resource utilization among library users at SMCC?

1.6 Hypothesis

Hypotheses

The following null hypotheses will be tested at 0.05 level of significance:

Ho_1: There is no significant relationship between the identified barriers and the level of online resource utilization among library users at SMCC.

Ho_2: There is no significant relationship between the identified enablers and the level of online resource utilization among library users at SMCC.

1.7 Scope and Delimitations of the Study

This study focuses on examining the barriers and enablers in the utilization of online resources among student users at Saint Michael College of Caraga (SMCC)**.** Specifically, it assesses the extent to which barriers such as lack of awareness, technological limitations, limited resource availability, and accessibility issues affect the use of online resources. Likewise, it explores the enablers that support utilization, including librarian support and instruction, faculty support, user training, and adequate infrastructure.

The study also evaluates how students utilize online resources in terms of frequency of access, purpose of use, level of satisfaction, and perceived effectiveness. Moreover, it examines the relationship between the identified barriers and enablers and the overall level of online resource utilization.

This research will be conducted during the School Year 2025–2026, covering both the first and second semesters to ensure the completion of all phases, including proposal defense, data gathering, data analysis, and final reporting. The respondents will be active student users of SMCC's online library resources. Their course and year level will only be used as demographic information to describe the respondent profile and will not be included as part of the study variables.

The scope of the study is limited to online resources officially provided by SMCC, such as e-books, e-journals, subscribed databases, institutional repositories, and other digital learning platforms. It excludes print materials, non-academic online sources, and external databases not officially subscribed to or managed by the institution.

Furthermore, the study does not include the perspectives of librarians, faculty members, or administrators, as it centers solely on student experiences. The findings will also be limited by the self-reported nature of the survey data, the geographic coverage confined to SMCC, and the time frame restricted to one academic year, which may affect the generalizability of the results to other institutions.

1.8 Definition of Terms

The following terms are operationally defined as they are used in this study:

Accessibility Issues – These refer to the difficulties encountered by library users in accessing online library resources, as measured by their responses regarding unstable internet connectivity, login restrictions, outdated systems, and device incompatibility in the survey questionnaire.

Adequate Infrastructure – This refers to the availability of functional internet connection, computers or personal devices, and related technological facilities, as perceived by library users and measured through survey items on infrastructure sufficiency.

Barriers – These are the factors that hinder library users from effectively utilizing online library resources, identified in this study through survey indicators such as lack of awareness, poor connectivity, limited infrastructure, and insufficient guidance.

Effectiveness of Use – This refers to the extent to which online library resources support users' academic tasks, as measured by self-reported improvements in research output, completion of assignments, and learning performance.

Enablers – These are the facilitating factors that promote the use of online library resources, measured through users' perceptions of librarian assistance, ICT training, faculty encouragement, and availability of reliable technology.

Faculty Support – This refers to the involvement of faculty members in promoting online resource use, measured through survey items related to course integration, research guidance, and encouragement provided to students.

Frequency of Access – This refers to how often library users access online library resources, measured in this study through self-reported usage frequency such as daily, weekly, or rarely.

Information Literacy – This refers to the ability of library users to identify, locate, evaluate, and use online information resources, as measured through survey items assessing users' confidence and skills in information handling.

Lack of Awareness – This refers to users' limited knowledge of the availability and functions of online library resources, measured through survey responses indicating unfamiliarity or insufficient orientation.

Librarian Instruction and Support – This refers to the assistance provided by librarians in the form of orientations, one-on-one guidance, and training sessions, as measured through users' perceived level of librarian support.

Level of Satisfaction – This refers to the degree of users' satisfaction with online library resources in terms of availability, relevance, and usefulness, measured using Likert-scale survey items.

Limited Availability of Resources – This refers to the insufficiency of accessible digital collections or subscription databases, as measured through users' perceptions of resource adequacy in meeting their academic needs.

E-Resources – These refer to electronic academic materials such as e-books, e-journals, online databases, and institutional repositories, as identified and accessed by users through academic libraries included in the study.

Online Resource Utilization – This refers to the actual use of online library resources by library users, operationalized in this study through frequency of access, purpose of use, perceived usefulness, and satisfaction level.

Purpose of Use – This refers to the specific academic reasons for accessing online library resources, measured through survey responses such as research activities, course assignments, and independent learning.

Technological Limitations – These refer to technology-related constraints such as slow internet connection, lack of ICT skills, and absence of appropriate devices, as measured through users' self-reported experiences.

User Training – This refers to formal orientations, workshops, or training activities provided to users, measured through respondents' reported participation and perceived usefulness of such training.

Library Users – These refer to students, faculty members, and researchers who utilize the services and online resources of academic libraries in Agusan del Norte, as identified as respondents of the study.

Academic Libraries – These refer to higher education institution libraries in Agusan del Norte that provide both physical and online resources, which serve as the setting of this research.



CHAPTER 2

RESEARCH METHODOLOGY

This chapter aims to give an overview that establishes a frame of reference for the study's entries and to synthesize the facts and information found in relevant literature and studies, both local and global, that are of great significance.

2.1 Research Design

This study will use a descriptive research design as a tool for assessing, collecting, and analyzing data to identify the barriers and enablers in online resource utilization among library users in Saint Michael College of Caraga (SMCC). The descriptive design will allow the researchers to describe current conditions and relationships among variables without manipulating them.

Furthermore, a quantitative research method will be employed to ensure objectivity, reliability, and validity of the findings. This approach will emphasize the use of measurable data and statistical analysis derived from survey questionnaires to determine patterns, relationships, and levels of utilization among respondents.

2.2 Research Materials and Methods

This study will utilize a modified survey questionnaire adapted from the studies of Fagbemi et al. (2021), *Online Information Resources Utilization among Students in Adeyemi College of Education, Ondo, Nigeria*, and Mole (2017), *Assessment of Academic Utilization of Online Information Resources by Undergraduate Students in the University of Nigeria, Nsukka.*

A structured questionnaire will be used to gather quantitative data from respondents regarding the barriers and enablers they experience in utilizing online library resources. The

results of the survey will serve as the basis for proposing recommendations to improve online resource utilization in academic libraries.

The questionnaire will be divided into four main sections:

Section I – Demographic Profile

This section will gather background information about the respondents, such as age, sex, year level, course or program, and library usage habits. These data will help determine how demographic characteristics relate to the utilization patterns of online resources.

Section II – Barriers to Online Resource Utilization

This section will focus on the obstacles that respondents may encounter while using online library facilities. Barriers may include lack of internet access, limited digital literacy, insufficient technical support, or lack of awareness of available e-resources. Respondents will rate each statement using a four-point Likert scale:
1 – Strongly Disagree, 2 – Disagree, 3 – Strongly Agree, 4 – Agree.

Section III – Enablers of Online Resource Utilization

This section will assess the factors that make the use of e-resources more convenient and accessible to respondents, such as library orientation, user-friendly platforms, librarian assistance, institutional support, and cost-effectiveness. A five-point Likert scale will also be used in this section.

Section IV – Level of Online Resource Utilization

This section will determine how respondents utilize online resources in terms of frequency of access, purpose of use, level of satisfaction, and effectiveness of utilization. Like the previous sections, this part will also use a five-point Likert scale to quantify responses.

2.3 Research Environment, Population and Sample

This study will be conducted in the Learning Resource Center of Saint Michael College of Caraga (SMCC) across different departments the College of Computing and Information Science (CCIS), College of Teacher Education Arts and Sciences (CTEAS), College of Business Management and Accountancy (CBMA), College of Tourism and Hospitality Management (CTHM), and College of Criminal Justice Education (CCJE) .

Research Locale



This study will be conducted at Saint Michael College of Caraga (SMCC), a higher education institution located in Barangay 4 Atupan Street, Nasipit, Agusan del Norte, Philippines. It is conveniently situated along Atupan Street, near Saint Michael the Archangel Parish Church, Versoza Park, and the Nasipit Municipal Hall. Adjacent to the rectory, the school site encompasses an area of 1,630 sq. m. In 2018, SMCC joined the Philippine Society for Quality and achieved the ISO 9001:2015 Surveillance Audit on February 7-8, 2019, demonstrating compliance with the Quality Management System. Currently, SMCC constructed a new building at the Triangulo Campus, with the aspiration of becoming a university by 2035. The institution envisions

providing top-notch education and training that meets the ever-evolving demands of today's world.

Research Respondents

The respondents of this study will be library users from the different departments above. Only active users who are willing to participate will be included. The stratified random sampling technique will be employed to ensure representation across the different departments. The exact number of respondents will be determined once permission for data collection is granted.

Instrument Validation

The survey questionnaire has been reviewed and validated by a statistician to ensure clarity, appropriateness, and suitability for the study. This ensures that the instrument is ready for accurate and reliable data collection.

2.4 Research Data and Analysis Plan

The respondents of this study are the users in the Learning Resource Center of SMCC across different departments College of Computing and Information Science (CCIS), College of Teacher Education Arts and Sciences (CTEAS), College of Business Management and Accountancy (CBMA), College of Tourism and Hospitality Management (CTHM), and College of Criminal Justice Education (CCJE) . The stratified random sampling technique will be used by the researchers.

Data Gathering Procedure

The researchers will prepare a formal letter addressed to the Vice President for Academic Affairs seeking approval to conduct the study. Upon approval, a request will be sent to the head librarian for permission to distribute the survey to eligible respondents.

Once the target population has been identified, the researchers will consult a statistician for advice on sampling strategy and statistical treatment to be used in data analysis. If an existing

standardized survey instrument is used, the researchers will secure permission from the original author(s) to ensure ethical use and proper implementation.

Data Analysis

Descriptive statistical techniques such as frequency, percentage, mean, and standard deviation will be employed to analyze the data and determine the extent of barriers and enablers, as well as the level of online resource utilization among library users at Saint Michael College of Caraga (SMCC). These statistical measures will be used to address Problems 1 to 3 of the study.

To examine the relationships between variables, inferential statistical analysis will be utilized. For Problem 4**,** which aims to determine the significant relationship between the identified barriers and the level of online resource utilization among library users at SMCC, the Pearson Product–Moment Correlation Coefficient (Pearson r) will be applied.

Likewise, for Problem 5**,** which seeks to determine the significant relationship between the identified enablers and the level of online resource utilization among library users at SMCC, the Pearson Product–Moment Correlation Coefficient (Pearson r) will also be used.

The use of Pearson r is appropriate since the variables are measured using Likert-scale instruments and treated as interval data. All inferential statistical tests will be conducted at a 0.05 level of significance. Statistical software will be utilized to ensure accurate and systematic data processing, and the results will be interpreted to identify significant patterns and relationships that will serve as the basis for drawing conclusions aligned with the objectives of the study.

2.5 Research Ethical Standards

The researchers strictly adhered to established ethical standards and research protocols to ensure the credibility, integrity, and reliability of the study. All procedures were conducted in accordance with institutional guidelines and ethical principles governing human participation in research.

A. Protection of Intellectual Property Rights (IPR)

All references, studies, and online materials used in this research were properly cited and acknowledged under the APA 7th Edition format. The researchers took every possible measure to ensure that the study did not violate the intellectual property rights of the authors, institutions, or organizations that were used. The researchers' original contributions were protected while others' intellectual property was respected by appropriately citing any third-party sources they used, such as secondary data or previously published works. As the study did not entail the development of a brand, logo, or sign, registration of a trademark was not relevant.

B. Informed Consent

Informed consent is given by each participant before they begin the survey. Before data collection, the participants were provided with complete information about the study's purpose, scope, and objectives. The participants were required to read and sign an informed consent form that indicated their voluntary agreement to participate in the research. The consent form explicitly stated that participation was voluntary and that refusal to participate would not result in any academic or institutional consequences.

C. Data Privacy and Confidentiality

This research strictly follows the Data Privacy Act of 2012 (RA 10173) in the Philippines. All data received from the participants throughout the entire process were considered confidential. Protecting the participant anonymity, personal identifications such as names, addresses, or any other contact details were excluded. Data were kept in encrypted and

password-protected files which were accessible only to the research team. A participant's consent and respect to legal and ethical criteria are requirements for sharing anonymized data with approved third parties for academic purposes.

D. Voluntary Participation and Freedom to withdraw

Participation in this study was strictly voluntary and free from any form of coercion or undue influence, particularly since the respondents included students and internal members of the institution. No participant was pressured, required, or compelled by instructors, administrators, librarians, or the researchers to take part in the study.

Participation or non-participation had no effect on students' academic standing, grades, class requirements, or access to library services, and no incentives, penalties, or rewards were associated with participation. Respondents were fully informed that they had the right to decline participation or withdraw from the study at any point without providing a reason and without experiencing any negative consequences.

E. Minimization of Harm and Risk Management

There were no physical, psychological, or social harms to the participants. To prevent any potential pain, the survey questions were designed to be non-intrusive. The risks that might have occurred during the data collection process were eliminated since such processes were done in a manner that respected the privacy and emotional well-being of the participants.

F. Beneficence and Contribution to Knowledge

This study aimed to significantly advance library and information science by determining the factors that either support or undermine the efficient use of internet resources. Because the findings will make it easier for students, librarians, and academic institutions to access and utilize online library resources, they are expected to be beneficial.

G. Justice and Fair Participant Selection

Participants will be selected based on fair and transparent inclusion and exclusion criteria. The respondents were selected according to the focus of the study - library users from certain academic libraries in Agusan del Norte. The selection process was not influenced by the respondents' sex, academic status, or socio-economic background, thus allowing every one of them to take part.

H. Data Integrity and Accuracy

When collecting, encoding, and analyzing data, the research team took care to be extremely accurate and truthful. Fake or manipulated data was completely unacceptable. All survey responses were objectively analyzed to ensure that reliable results were obtained, and to preserve credibility and transparency.

I. Transparency and Honesty in Reporting

There will be no data manipulation, and all research findings will be presented honestly. Then also identified limitations that may affect how the findings are interpreted. All third-party items, including literature or tools, shall be appropriately cited to respect intellectual property rights, and any potential conflicts of interest will be addressed.

J. Use of Patented or Copyrighted Materials

Prior authorization from the legitimate owners or authors must be obtained if copyrighted or patented items, such as published literature and secondary sources, were used. There was appropriate recognition in the references section. Proper citations will be included in the final report to acknowledge these sources and ensure compliance with copyright laws.

K. Ethical Considerations for Animal and Human Trials

The study was limited to human participants (library users) through surveys, there was no animal testing or experimentation done. The involvement of respondents was done following the ethical research standards of respect and voluntary participation.

L. Responsible Use of AI and Other Related Technologies

Digital technology and artificial intelligence tools will be used in the study to support different phases of the research process. Throughout the preliminary analysis stage, ChatGPT will be used to arrange ideas and compile preliminary results. Grammarly will make sure that the research paper followed correct grammar and clarity guidelines. Assuring that no AI tools contribute to bias or misinterpretation and evaluating AI-generated outputs are essential components of ethical usage of these technologies.

M. Ethical Clearance and Institutional Approval

Prior to data collection, the Research Ethics Committee of Saint Michael College of Caraga (SMCC) granted its ethical clearance and approval, ensuring it meets guidelines for participant protection, data privacy, and intellectual property rights. Only once the institution and collaborating libraries provided the necessary authorization could the study begin.

2.6 Bibliography

Abiodun Alasa, S., & Oluwaseyi Quadri, G. (2022). E-resources usage among polytechnic students in Southwest Nigeria: evidence from Federal Polytechnic, Ede and The Polytechnic, Ibadan Nigeria. *International Journal of Knowledge Content Development & Technology*, *12*(1). https://doi.org/10.5865/IJKCT.2022.12.1.049

Abuhassna, H., Al-Rahmi, W. M., Yahya, N., Zakaria, M. A. Z. M., Kosnin, A. B. M., & Darwish, M. (2020). Development of a new model on utilizing online learning platforms to improve students' academic achievements and satisfaction. *International Journal of Educational Technology in Higher Education*, *17*(1). https://doi.org/10.1186/s41239-020-00216-z

Adeyeye, B., Ojih, S. E., Bello, D., Adesina, E., Yartey, D., Ben-Enukora, C., & Adeyeye, Q. (2022). Online learning platforms and covenant university students' academic performance in practical related courses during COVID-19 Pandemic. *Sustainability (Switzerland)*, *14*(2). https://doi.org/10.3390/su14020878

Al-Khatib, M., Alkhatib, A., Talhami, M., Kashem, A. H. M., Ayari, M. A., & Choe, P. (2024). Enhancing engineering students' satisfaction with online learning: factors, framework, and strategies. *Frontiers in Education*, *9*. https://doi.org/10.3389/feduc.2024.1445885

Aryee, R., & Apronti Tetteh, E. O. (2024). Library resources constraints, frustration, and user behavior: An empirical library operations study. *Brain and Behavior*, *14*(7). https://doi.org/10.1002/brb3.3627

Azrael, T., & Florentino, P. (n.d.-a). Level of awareness and effectiveness of open access resources among library users at lyceum of the Philippines University. *Issue 1 Information Generation & Management*, *4*(1), 44–74. https://doi.org/10.5281/zenodo.15321595

Azrael, T., & Florentino, P. (n.d.-b). Level of awareness and effectiveness of open access resources among library users at Lyceum of the Philippines University. *Issue 1 Information Generation & Management*, *4*(1), 44–74. https://doi.org/10.5281/zenodo.15321595

Basar, Z. M., Mansor, A. N., Jamaludin, K. A., & Alias, B. S. (2021). The effectiveness and challenges of online learning for secondary school students - a case study. *Asian Journal of University Education*, *17*(3), 119–129. https://doi.org/10.24191/ajue.v17i3.14514

Becksford, L. (2022). Instruction librarians' perceptions of the faculty–librarian relationship. *Communications in Information Literacy*, *16*(2), 119–150. https://doi.org/10.15760/comminfolit.2022.16.2.3

Briones, M. R., Prudente, M., & Errabo, D. D. (2023). Characteristics of filipino online learners: a survey of science education students' engagement, self-regulation, and self-efficacy. *Education Sciences*, *13*(11). https://doi.org/10.3390/educsci13111131

Burhansab, P. A., Sadik Batcha, M., Ahmad, M., & Batcha, S. (n.d.). *Investigating Awareness and usage of Electronic Resources by the Library Users of Selected Colleges of Solapur University*. https://digitalcommons.unl.edu/libphilprac

Buruga, B. A., Ali, G., & Izaruku, R. (2024). Comprehensive assessment of drivers and barriers to electronic information resources usage in academic libraries in uganda using SWOT analysis. *IP Indian Journal of Library Science and Information Technology*, *8*(2), 113–129. https://doi.org/10.18231/j.ijlsit.2023.019

Chesi, E. S., Kongnyuy, P., & Jehovah, C. L. O. (2025). The effect of mentorship on pre-service teachers' perceived competence in addressing diverse learners' needs In Mezam Division, Cameroon. *European Journal of Education Studies*, *12*(9). https://doi.org/10.46827/ejes.v12i9.6174

Colbongan-Papay, L. C., Colas, J. B., Papay, L. C., & Colas, J. B. (n.d.). *Status of Blended Library Services at the University of Baguio*. https://digitalcommons.unl.edu/libphilprac

Cunningham, Paul., & Cunningham, Miriam. (2020). *2020 IST-Africa Conference : 18-22 May 2020, Virtual Conference*. IEEE.

Daramola, F. O. (2022). Utilization of e-resources for learning among undergraduates. *Indonesian Journal of Educational Research and Technology*, *3*(1), 35–44. https://doi.org/10.17509/ijert.v3i1.44731

Defrain, E., Sult, L., & Pagowsky, N. F. (n.d.). *Effectiveness of academic library research guides for building effectiveness of academic library research guides for building college students' information literacy skills: a scoping review college students' information literacy skills: a scoping review academic library guides and learning effectiveness studies 1 effectiveness of academic library research guides for building college students' information literacy skills: A scoping review introduction*. https://digitalcommons.unl.edu/libraryscience

Diki, P. (2022). *Factors influencing utilization of library e-resources in academic delivery: the case of multimedia university of kenya background of the study*. http://journal.kyu.ac.ke/index.php/library1http://journal.kyu.ac.ke/index.php/library1

Durodolu, O. O., Iroaganachi, M. A., & Otunla, J. N. (2024). Effects of slow internet on academic library staff productivity and job satisfaction 1 2 3. *Berkala Ilmu Perpustakaan Dan Informasi*, *20*(2), 435–447. https://doi.org/10.22146/bip.v19i1.14705

Ec, P. (2023). *Assessment and utilization of digital library services in the new normal: implications to students' academic application*. *6*, 111–123. https://doi.org/10.17501/26307413.2023.6109

Efe Okolo, S., & Efe Stanley, O. (n.d.-a). *Utilization of electronic resources for effective utilization of electronic resources for effective library services in the 21st century library services in the 21st century utilization of electronic resources for effective library services in the 21 st century*. https://digitalcommons.unl.edu/libphilprac

Efe Okolo, S., & Efe Stanley, O. (n.d.-b). *Utilization of electronic resources for effective utilization of electronic resources for effective library services in the 21st century library services in the 21st century utilization of electronic resources for effective library services in the 21 st century*. https://digitalcommons.unl.edu/libphilprac

EPRA International Journal of Environmental Economics, Commerce and Educational Management. (n.d.). https://doi.org/10.36713/epra0414

Estrada, C. Y. E., Gutierrez, F. V. O., Valdez, M. G., & Alenzuela, R. C. (2023). Enhancing academic success through utilization of library resources in a public University in the Philippines. *Asia Pacific Journal of Social and Behavioral Sciences*, *21*, 71–83. https://doi.org/10.57200/apjsbs.v21i0.351

Fabito, B., Trillanes, A., & Sarmiento, J. (2021). Barriers and challenges of computing students in an online learning environment: insights from one private university in the philippines.

International Journal of Computing Sciences Research, *5*(1), 441–458. https://doi.org/10.25147/ijcsr.2017.001.1.51

Fagbemi, V. Y., Awoyemi, R. A., & Olubiyo, P. O. (n.d.). *Online information resources utilization among Students in Adeyemi College of Education, Ondo, Nigeria*.

Faix, A., & Baker, A. (n.d.). *Create engaging library resource lists: helping faculty discover create engaging library resource lists: helping faculty discover educational technologies for curating and sharing library educational technologies for curating and sharing library resources resources*. https://digitalcommons.kennesaw.edu/seln

Hazarika, K. (2024). *Exploring the awareness and usage of electronic library exploring the awareness and usage of electronic library resources among students at college level in assam resources among students at college level in Assam*. https://digitalcommons.unl.edu/libphilprac

Humbhi University Of Balochistan, S., Shabbir, P., Law, T., Affairs, P., Balochistan, D., Raheem, P. A., & Humbhi, A. (n.d.). *Awareness and utilization of e-resources to support academic and research activities.Impact-of-Library-Instruction-for-Effective-Utilization-of-Library-Resources-by-Fresh-Students-of-Sokoto-State-University-Sokoto-Nigeria*. (n.d.).

Jameson, J., Natal, G., & Napp, J. (2019). *Evolving and Enduring Patterns Surrounding Student Usage and Perceptions of Academic Library Reference Services* (Vol. 80, Issue 3). https://crl.acrl.org/index.php/crl/

Kayode, O. V., Onafowope, ;, Adesola, M., Oweikpodor, ;, Gbaeprekumo, V., Obona, ;, Egbe, E., & Id, O. (n.d.). *Utilization of digital resources and academic staff job effectiveness in university of Calabar, Nigeria*.

Lawal, M. T., & Kannan, S. (2020). Availability of information resources, services, facilities and utilization as a determining factor for user satisfaction among students of federal university of agriculture abeokuta ogun state nigeria. *International Journal of Research in Library Science*, *6*(1), 53. https://doi.org/10.26761/ijrls.6.1.2020.1304

Lu, Y., & Lin, S. (2024). Digital transformation in college libraries: The effect of digital reading on reader service satisfaction. *PLoS ONE*, *19*(8). https://doi.org/10.1371/journal.pone.0307699

Lucero, H., Victoriano, J., Carpio, J., & Fernando, P. Jr. (2021). assessment of e-learning readiness of faculty members and students in the government and private higher education institutions in the Philippines. *International Journal of Computing Sciences Research*, *5*(1), 398–406. https://doi.org/10.25147/ijcsr.2017.001.1.48

Mae Toroba, R. S., & Tagadiad, C. L. (n.d.). *The mediating effect of self-regulation on the relationship between self-directed learning and digital competence*. https://doi.org/10.51584/IJRIAS

Medicherla, V. S., Sadik Batcha, M., Naga, Y. M. V, & Rao, R. (n.d.). (2024) An empirical study on assessing library resource utilization and service impact in degree colleges of west godavari district. In *Library Progress International|* (Vol. 44, Issue 3). www.bpasjournals.com

Musungu, M. S., & Maina, J. (2023). Accessibility and utilization of information resources in the post covid-19 era: a case of selected university libraries in kenya. In *KLISC Journal of Information Science and Knowledge Management* (Vol. 1, Issue 1). https://orcid.org/0000-0001-5925-3625

Mwantimwa, K., Mwabungulu, E., & Kassim, M. (2021). Academic staff and researchers' use of electronic resources in tanzania: a comparative study. In *International Journal of Education and Development using Information and Communication Technology (IJEDICT)* (Vol. 17, Issue 2).

Nguyen, T. L., & Tuamsuk, K. (2021). Faculty–librarian administrative structure and collaborative activities supporting teaching and research at Vietnamese universities: A qualitative study. *IFLA Journal*, *47*(2), 236–249. https://doi.org/10.1177/0340035220944945

Norch, C. K., & Adzakpa, K. A. (2022). Assessing the Usage of Library Electronic Resources by Postgraduate Students at the University of Ghana Balme Library Research Common. *OALib*, *09*(09), 1–17. https://doi.org/10.4236/oalib.1109187

OKEKE Senior Librarian, J., & Amaechi, N. (2019). The use of e-library resources as a correlate of user satisfaction in university libraries in Anambra State, Nigeria. *In International Journal of Social Sciences and Humanities Reviews* (Vol. 9, Issue 1).

Panhilason, A. (2024a). Users' awareness, utilization, and satisfaction with online library information resources and services. *Journal of Interdisciplinary Perspectives*, *2*(3). https://doi.org/10.69569/jip.2024.0034

Panhilason, A. (2024b). Users' awareness, utilization, and satisfaction with online library information resources and services. *Journal of Interdisciplinary Perspectives*, *2*(3). https://doi.org/10.69569/jip.2024.0034

Raihanatu Hamid, M., Aji, S., Ajuji, M., & Ibrahim, A. (n.d.). *assessment of accessibility and utilization of internet services among staff and students of college of education Billiri*. www.ijmrsti.com

Roman, R. G. (2020). A study on the utilization of e-resources among college students. *International Journal of Knowledge Engineering*, 24–29. https://doi.org/10.18178/ijke.2020.6.1.127

Rosales, A. C. (n.d.). *An empirical study on knowledge, access and utilization of electronic resources by University Students in Philippines*. https://doi.org/10.47772/IJRISS

Rosales, A. C. (2024). An empirical study on knowledge, access and utilization of electronic resources by university students in philippines. *International Journal of Research and Innovation in Social Science*, *VIII*(V), 23–42. https://doi.org/10.47772/ijriss.2024.805003

Shikali, J. C., Muneja, P. S., & Kassim, M. (n.d.). *Access to Library Information Resources by University Students during COVID-19 Pandemic in Africa: A Systematic Literature Review*.

Singh, M. K. (n.d.). Awareness and use of e-resources among the users of library of punjabi university patiala: a case study. In *Journal of Indian Library Association* (Vol. 55, Issue 4).

Spangler, S. C., & Casper, D. (2023). Faculty's perceptions of embedded librarians. *Issues in Information Systems*, *24*(4), 145–160. https://doi.org/10.48009/4_iis_2023_112

Susanti, E. R., Suryati, N., & Astuti, U. P. (2021). *Students' perception on the utilization of learningapps.org for self-study materials*.

Technology-Enhanced Self-Study Learning Tool_ An Approach __to Improve Students Skills. (n.d.).

Tong, D. H., Uyen, B. P., & Ngan, L. K. (2022). The effectiveness of blended learning on students' academic achievement, self-study skills and learning attitudes: a quasi-experiment study in teaching the conventions for coordinates in the plane. *Heliyon*, *8*(12). https://doi.org/10.1016/j.heliyon.2022.e12657

Tukur, L. M. (2019). *Issues and challenges in accessing and utilization of information resources by students of FUAA*. https://digitalcommons.unl.edu/libphilprac

Usman, K., & Younus, M. (2023). Examine the challenges and barriers hindering distance learners' access to academic library resources. *Journal of Education and Social Studies*, *4*(3), 813–820. https://doi.org/10.52223/jess.2023.4342

Vanessa, M., Teofilo-Labitad, C., & Paglinawan, J. L. (n.d.). *The relationship between digital resource availability and intentional learning among high school science students in Valencia City, Bukidnon, Philippines*. https://doi.org/10.47772/IJRISS

(*Document (3)*, n.d.)

CHAPTER 3

RESULTS AND DISCUSSION

3.1 Presentation and Analysis of Data

3.2 Proposed Intervention/ Output

DELETE BOX AFTER READING

Check the guidelines in research writing.

Make sure to integrate Mendeley in the citation.

Use Calibri 11 and double space

APPENDICES

Survey Questionnaire

Survey Questionnaire

Title:

Survey Questionnaire on Barriers and Enablers in Online Resource Utilization among Library Users in Saint Michael College of Caraga (SMCC)

Purpose:

This questionnaire aims to gather data on the barriers, enablers, and utilization patterns of online library resources among library users in SMCC.

Instruction:

Please indicate your level of agreement with each statement by checking (✔) the box that best represents your opinion.

(5 – Strongly Agree, 4 – Agree, 3 – Neutral, 2 – Disagree, 1 – Strongly Disagree)

Part I. Respondents' Profile

1. Name: (Optional)
2. Sex: ☐ Male ☐ Female
3. Age: _____
4. Type of Department: ☐ CCIS ☐ CBMA ☐ CTEAS ☐ CCJE ☐ CTHM
5. Year Level: ☐ 1st Year ☐ 2nd Year ☐ 3rd Year ☐ 4th Year
6. Type of Institution: ☐ Public ☐ Private
7. Frequency of Using Online Library Resources:
   ☐ Daily ☐ Weekly ☐ Monthly ☐ Rarely
8. Type of Device Used for Access: ☐ Smartphone ☐ Laptop ☐ Desktop ☐ Tablet

Part II. Barriers Affecting Online Resource Utilization

A. Lack of Awareness

(adapted from Olubiyo et al., 2021; Mole, 2017)

Statements	**5**	**4**	**3**	**2**	**1**
1. I am not fully aware of the online resources available in our library.	☐	☐	☐	☐	☐
2. I rarely receive updates or announcements about new online resources.	☐	☐	☐	☐	☐
3. I do not know how to search or locate specific online databases.	☐	☐	☐	☐	☐
4. The library provides limited orientation or training about its online collections.	☐	☐	☐	☐	☐

B. Technological Limitations

Statements	**5**	**4**	**3**	**2**	**1**
5. Unstable internet connection affects my access to online resources.	☐	☐	☐	☐	☐
6. Power interruptions make it difficult to use online library platforms.	☐	☐	☐	☐	☐
7. The online library website or database loads slowly.	☐	☐	☐	☐	☐
8. There are not enough computers or gadgets available for use in the library.	☐	☐	☐	☐	☐

C. Limited Resource Availability

Statements	**5**	**4**	**3**	**2**	**1**
9. Some online databases subscribed by the library are outdated or limited.	☐	☐	☐	☐	☐



10. The library's online resources do not cover all subjects I need.	☐ ☐ ☐ ☐ ☐
11. Some e-books or journals are restricted and cannot be accessed freely.	☐ ☐ ☐ ☐ ☐
12. I often fail to find relevant materials for my coursework.	☐ ☐ ☐ ☐ ☐

D. Accessibility Issues

Statements	**5 4 3 2 1**
13. The online library system is difficult to navigate.	☐ ☐ ☐ ☐ ☐
14. Login and authentication issues prevent me from accessing databases.	☐ ☐ ☐ ☐ ☐
15. The interface of the online library platform is not user-friendly.	☐ ☐ ☐ ☐ ☐
16. I experience difficulties accessing online resources outside the campus.	☐ ☐ ☐ ☐ ☐

Part III. Enablers Supporting Online Resource Utilization

A. Librarian Support and Instruction

Statements	**5 4 3 2 1**



17. Librarians provide assistance in locating online materials. ☐ ☐ ☐ ☐ ☐

18. The library staff conducts orientations or tutorials about online resources. ☐ ☐ ☐ ☐ ☐

19. I can easily ask librarians for help when I encounter difficulties online. ☐ ☐ ☐ ☐ ☐

20. The library actively promotes online resources through posters or social media ☐ ☐ ☐ ☐ ☐

B. Faculty Support

Statements	**5**	**4**	**3**	**2**	**1**
21. My instructors encourage us to use online library resources for assignments.	☐	☐	☐	☐	☐
22. Faculty members recommend reliable online sources for academic tasks.	☐	☐	☐	☐	☐
23. Professors provide guidance on how to properly cite online materials.	☐	☐	☐	☐	☐
24. Our course requirements promote the use of library online databases.	☐	☐	☐	☐	☐

C. User Training

Statements	**5**	**4**	**3**	**2**	**1**
25. I have attended library training sessions on digital or online literacy.	☐	☐	☐	☐	☐
26. I know how to evaluate the credibility of online information sources.	☐	☐	☐	☐	☐
27. The library offers sufficient tutorials on effective online searching.	☐	☐	☐	☐	☐
28. I am confident in using online library platforms without assistance.	☐	☐	☐	☐	☐

D. Adequate Infrastructure

Statements	**5**	**4**	**3**	**2**	**1**
29. The library has enough functional computers and internet access points.	☐	☐	☐	☐	☐
30. Online databases are regularly updated and properly maintained.	☐	☐	☐	☐	☐
31. The institution ensures stable power supply and connectivity in the library.	☐	☐	☐	☐	☐
32. There is an efficient technical support system for online resource access.	☐	☐	☐	☐	☐

Part IV. Level of Online Resource Utilization

A. Frequency of Access

Statements	**5**	**4**	**3**	**2**	**1**
33. I frequently access the library's online databases for research purposes.	☐	☐	☐	☐	☐
34. I use online library resources both inside and outside the campus.	☐	☐	☐	☐	☐
35. I visit the library website more often than I use printed materials.	☐	☐	☐	☐	☐
36. I regularly check the library's online Resources for new or updated materials.	☐	☐	☐	☐	☐

B. Purpose of Use

Statements	**5**	**4**	**3**	**2**	**1**
37. I use online resources mainly for academic assignments and research.	☐	☐	☐	☐	☐



38. I use online databases to stay updated with new information in my field.	☐	☐ ☐ ☐ ☐
39. I use online resources to prepare reports, projects, or presentations.	☐	☐ ☐ ☐ ☐
40. I use e-books or e-journals for exam preparation.	☐	☐ ☐ ☐ ☐

C. Level of Satisfaction

Statements	**5 4 3 2 1**
41. The quality of online materials from the library meets my academic needs.	☐ ☐ ☐ ☐ ☐
42. The online library platforms are easy to use and reliable.	☐ ☐ ☐ ☐ ☐
43. I am satisfied with the accessibility and availability of online resources.	☐ ☐ ☐ ☐ ☐
44. I am satisfied with the assistance provided by librarians regarding online access.	☐ ☐ ☐ ☐ ☐

D. Effectiveness of Utilization

Statements	**5 4 3 2 1**
45. Using online library resources helps improve the quality of my academic work.	☐ ☐ ☐ ☐ ☐
46. Online resources save me time and effort compared to printed materials.	☐ ☐ ☐ ☐ ☐

47. I can complete my academic tasks more efficiently using online databases. ☐ ☐ ☐ ☐ ☐

48. I feel more independent and confident in doing research using online resources. ☐ ☐ ☐ ☐ ☐

Likert Scale Interpretation

Range	**Description**
4.20 – 5.00	Strongly Agree
3.40 – 4.19	Agree
2.60 – 3.39	Neutral
1.80 – 2.59	Disagree
1.00 – 1.79	Strongly Disagree

Source of Instrument Design

Adapted and modified from:

- Olubiyo, P. O., Fagbemi, V. Y., & Awoyemi, R. A. (2021). *Online Information Resources Utilization among Students in Adeyemi College of Education, Ondo, Nigeria. Library Philosophy and Practice (e-journal), 5759.*
- Mole, A. J. C. (2017). *Assessment of Academic Utilization of Online Information Resources by Undergraduate Students in University of Nigeria, Nsukka. International Journal of Knowledge Content Development & Technology*, 7(3), 29–48.

DELETE BOX AFTER READING

Check the guidelines in research writing.

Make sure to integrate Mendeley in the citation.

Use Calibri 11 and double space

CHAPTER 4

SUMMARY, CONCLUSION, RECOMMENDATION,

4.1. Summary

4.2. Conclusion

4.3.Recommendation



REFERENCES

DELETE BOX AFTER READING

Check the guidelines in research writing.

Make sure to integrate Mendeley in the citation.

References is single space- Calibri 11

CURRICULUM VITAE

NAME: Abegail Abunda

ADDRESS: P-7, Tollbridge, Buenavista, Agusan del Norte

EMAIL: abegail_abunda@smccnasipit.edu.ph

Personal Information

Sex : Female

Date of birth : September 19, 2005

Height : 149cm

Weight : 47 kg

Civil Status : Single

Educational Attainment

Elementary : Buenavista Central Elementary School

Secondary : Saint James High School

Tertiary : Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD
N/A	**N/A**

CURRICULUM VITAE



NAME: Jay-uanie Mansartihan Binongcasan

ADDRESS: P-8, Lomboyan,Guinabsan, Buenavista Agusan Del Norte

EMAIL: jay-uanie_binongcasan@smccnasipit.edu.ph

Personal Information

Sex : Female

Date of birth : June 24, 2003

Height : 156 cm

Weight : 56 kg

Civil Status : Single

Educational Attainment

Elementary : Lomboyan Elementary School

Secondary : Guinabsan National High School

Tertiary : Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD
N/A	**N/A**

CURRICULUM VITAE



NAME: Michelle Lamoste Duarte

ADDRESS: D-1, Sacol, Buenavista, Agusan del Norte

EMAIL: michelleduarte2005@gmail.com

Personal Information

Sex : Female

Date of birth : December 09, 2005

Height : 145 cm

Weight : 48 kg

Civil Status : Single

Educational Attainment

Elementary : Sacol Elementary School

Secondary : Buenavista Special Education High School

Tertiary : Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD
N/A	**N/A**

CURRICULUM VITAE



NAME: Mariel Comendador Ordiz

ADDRESS: P-4, Brgy. Macalang, Buenavista, Agusan del Norte

EMAIL: marielordiz344@gmail.com

Personal Information

Sex : Female

Date of birth : March 08, 2005

Height : 144 cm

Weight : 41 kg

Civil Status : Single

Educational Attainment

Elementary : Buenavista Central Elementary School

Secondary : Buenavista Special Education High School

Tertiary : Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD
N/A	**N/A**

CURRICULUM VITAE



NAME: Alexander Poriol

ADDRESS: P-4, Sampinit Malapong, Buenavista Agusan del Norte

EMAIL: alexander_poriol@smccnasipit.edu.ph

Personal Information

Sex : Male

Date of birth : January 05, 2005

Height : 166 cm

Weight : 69 kg

Civil Status : Single

Educational Attainment

Elementary : Malapong Elementary School

Secondary : Buenavista National High School

Tertiary : Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD
Future Librarians and Information Scientists'	**Prince Charming**

Society	

APPENDICES

A - Letter of Approval with Certificate of Appearance and Photo

B - Instrument with Informed Consent

C - Map of the Research Locale *(Include Coordinates)*

D- Narrative & Photo Documentation

E- Ethics Committee Approval



CERTIFICATE OF TECHNOLOGY ASSESSMENT

CERTIFICATE OF AUTHENTIC AUTHORSHIP

This is to certify that I/we, the undersigned author(s), have solely and honestly written the entire paper titled: "Click here to enter text." All contents, ideas, data, and analysis presented in this paper are the results of my/our own diligent work and intellectual effort. The paper has been crafted to the best of my/our abilities, in full compliance with academic integrity standards and ethical research practices.

I/We further declare that:

1. This paper is free from any form of plagiarism or copyright infringement.
2. All referenced materials have been properly cited, and due credit has been given to the original authors and sources.
3. The work does not contain any falsified or fabricated data or information.

I/We acknowledge that if any rules or laws related to copyright, plagiarism, or other related legal or ethical standards are violated, Saint Michael College of Caraga shall not be held accountable for my/our actions. Any legal, academic, or ethical consequences arising from such violations shall solely be my/our responsibility.

Signed this

Date: Click here to enter text.

Location: Click here to enter text.

Author(s):

Click here to enter text.

Author

Click here to enter text.

Author

Click here to enter text.

Author

Click here to enter text.

Author